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ST ANDREW'S JUNIOR COLLEGE



Preliminary Examinations

Chemistry

8873/2

Higher 1

12 Sep 2018

Paper 2

0800 – 1000

Additional Materials: Data Booklet

READ THESE INSTRUCTIONS FIRST

Write your name and civics group on all the work you hand in.

Write in dark blue or black pen.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A:

Answer **ALL** the questions in this section in the spaces provided.

Section B:

Answer **ONE** question from in this section in the spaces provided.

You are reminded of the need for good English and clear presentation in your answers. The number of marks is given in brackets [] at the end of each question or part question.

For Examiners use only:

Section A		Section B	
Question	Marks	Question	Marks
1	16	1/2	20
2	29		
3	15		
Total	60	Total	20
TOTAL (Section A + Section B)			80

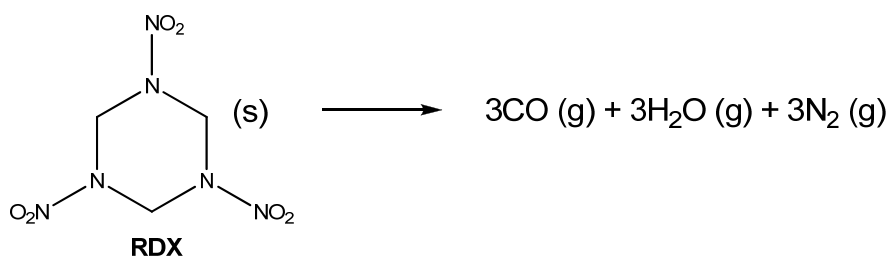
This document consists of **26** pages including a blank page.

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Section A

Answer **all** the questions in this section, in the spaces provided.

- 1 The explosive RDX is a white crystalline solid first prepared in 1899 but wasn't used as an explosive until 1920 by a German chemist called Herz. He prepared RDX by direct nitration of hexamine and found it be very sensitive to denotation by impact or friction. During World War 2, it was added to TNT to increase its explosive power. The explosive decomposition of RDX is as shown by the equation below.



- (a) (i) Calculate the volume of gases, in cm³, produced at s.t.p during the explosive decomposition of 2.22 g of RDX given that the relative molecular mass of RDX is 222.

[2]

- (ii) Write an equation to represent the standard enthalpy change of formation of RDX.

..... [1]

- (iii) Use the enthalpies of formation shown below to calculate the standard enthalpy change for the explosive decomposition of RDX.

Compound	$\Delta H_f / \text{kJ mol}^{-1}$
RDX (s)	+ 62
CO (g)	- 111
H ₂ O (g)	- 242

[2]

[Turn Over]

- 1 (a) (iv) Scientific literature usually quote the enthalpy change of formation for chemical explosives in kJ kg^{-1} . Calculate the enthalpy change of formation of RDX from kJ mol^{-1} to kJ kg^{-1} .

[1]

- (v) With reference to your answer in (iii), suggest if RDX is more stable than the products of decomposition.

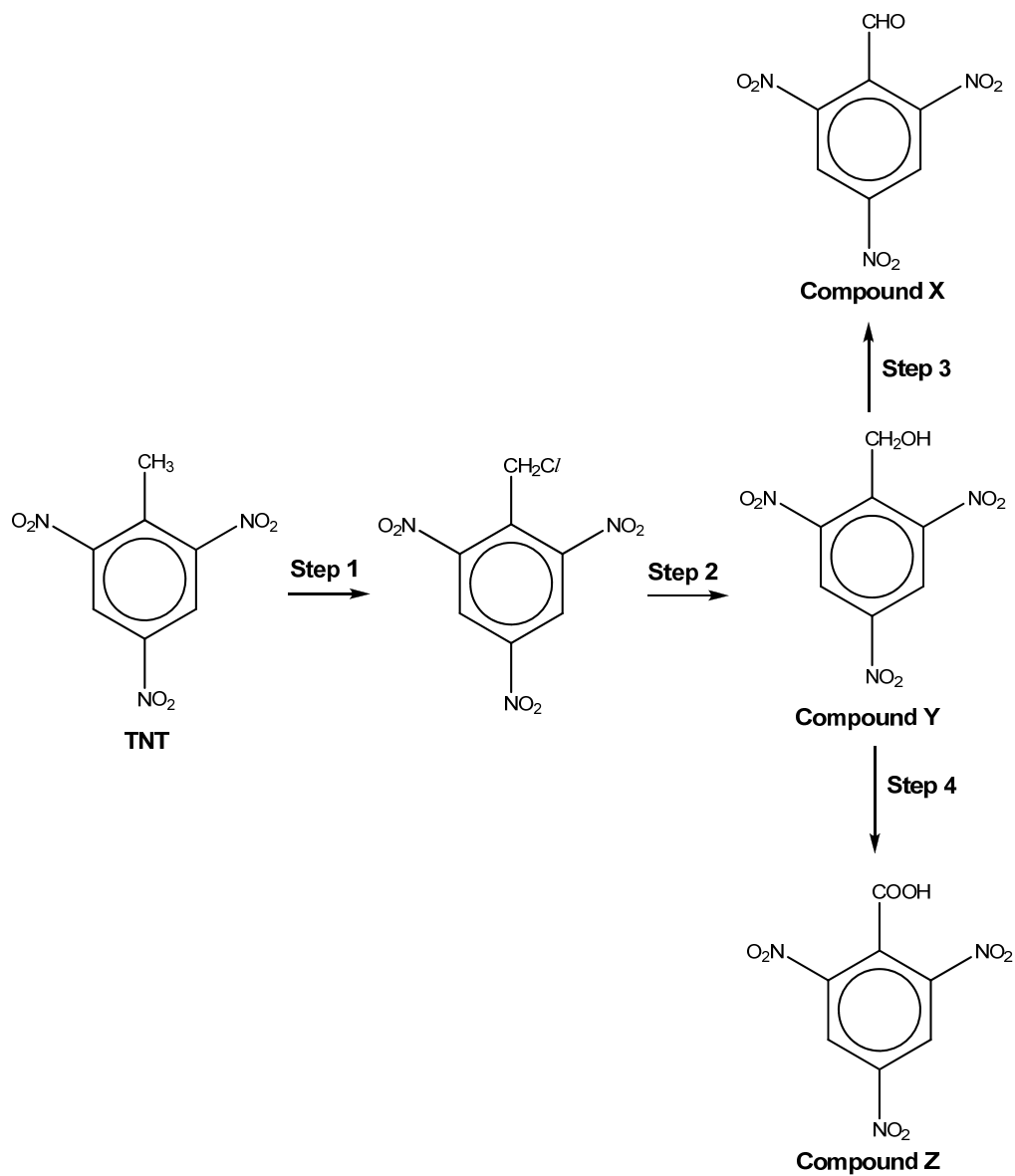
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[1]

[Turn Over

1 (b) The reaction scheme below shows the reactions involving TNT.



(i) State the reagent and condition for Steps 1 – 4.

Step 1:

Step 2:

Step 3:

Step 4:

[4]

[Turn Over

- 1 (b) (ii) Besides lithium aluminium hydride, state the reagent and condition that can reduce compound **X** into compound **Y**. Write a balanced equation for this reaction.

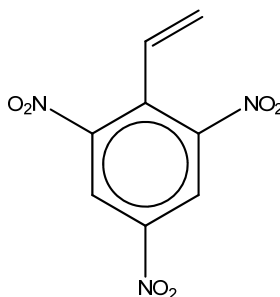
Reagent and condition:

Balanced equation:

.....

[2]

- (iii) Compound **Z** can also be synthesised by the oxidation of the following compound **W**.



Compound **W**

State and explain if the above compound exhibits cis-trans isomerism.

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[1]

- (iv) Suggest a suitable chemical test to distinguish between Compound **W** and TNT.

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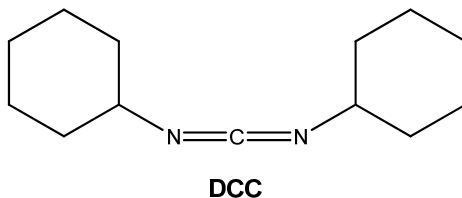
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[2]

[Total: 16]

[Turn Over

- 2 N,N-dicyclohexylcarbodiimide often abbreviated as DCC, is used to couple amino acids during artificial peptide synthesis. The structure of DCC is as shown below.



- (a) (i) Define the term *relative molecular mass*.

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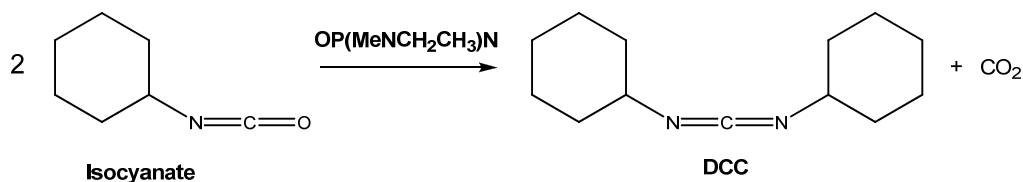
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[1]

- (ii) Calculate the relative molecular mass of DCC to 1 decimal place.

[1]

- (b) DCC can also be prepared by reacting isocyanate in the presence of a catalyst, $\text{OP}(\text{MeNCH}_2\text{CH}_3)_3$, as shown by the following equation. The yield of the reaction is typically 92.0%.



- (i) A student prepared DCC via the above synthesis method by using 0.100 g of isocyanate and 0.035 g of DCC was produced. Calculate the percentage yield of his experiment and hence suggest if the student's experimental technique is accurate given that the M_r of isocyanate is 125.0.

[2]

[Turn Over

- 2 (b) (ii) Explain, with the aid of an energy distribution diagram, how the addition of $\text{OP}(\text{MeNCH}_2\text{CH}_3)_3\text{N}$ affects the rate of the above chemical reaction.

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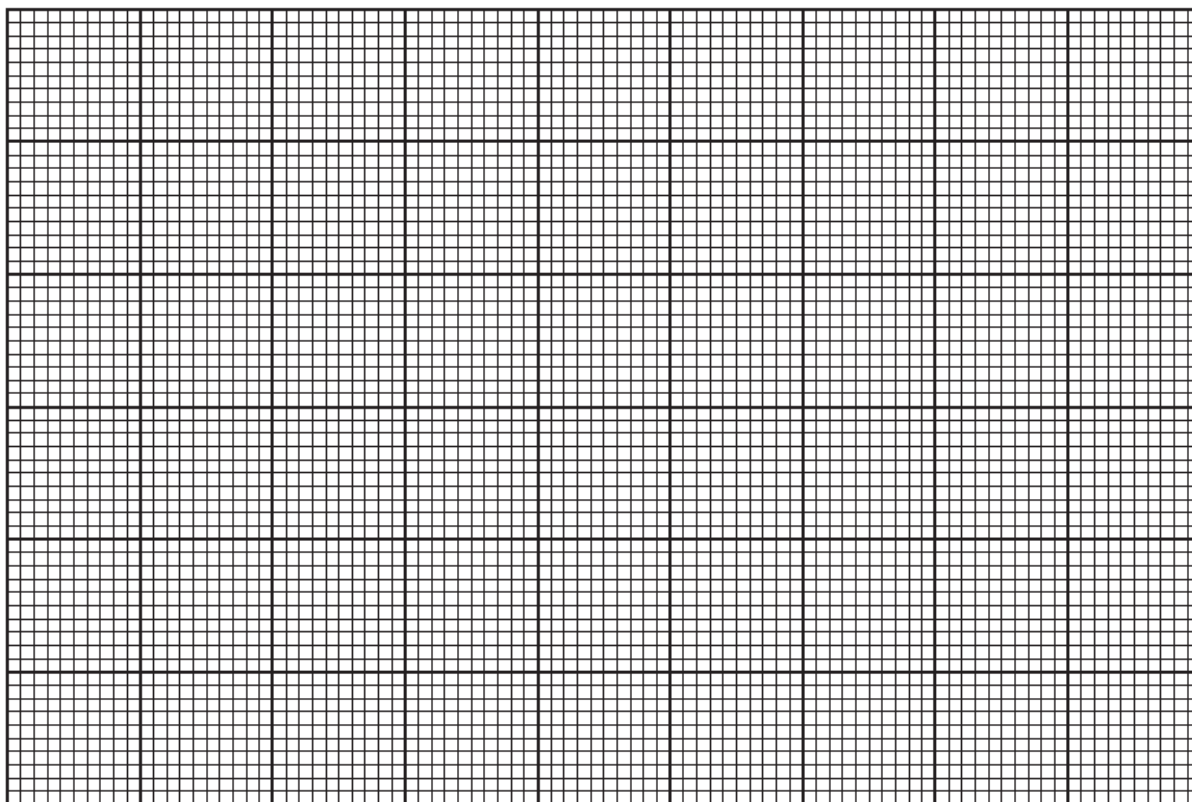
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[3]

[Turn Over

- 2 (b) (iii) The progress of the reaction can be followed by plotting the concentration of isocyanate against time. Plot the experimental results on the grid below to determine the rate equation for this reaction.

Time/ min	[Isocyanate] / mol dm ⁻³
0	0.0100
40	0.0079
80	0.0062
120	0.0049
160	0.0038
200	0.0030
240	0.0024



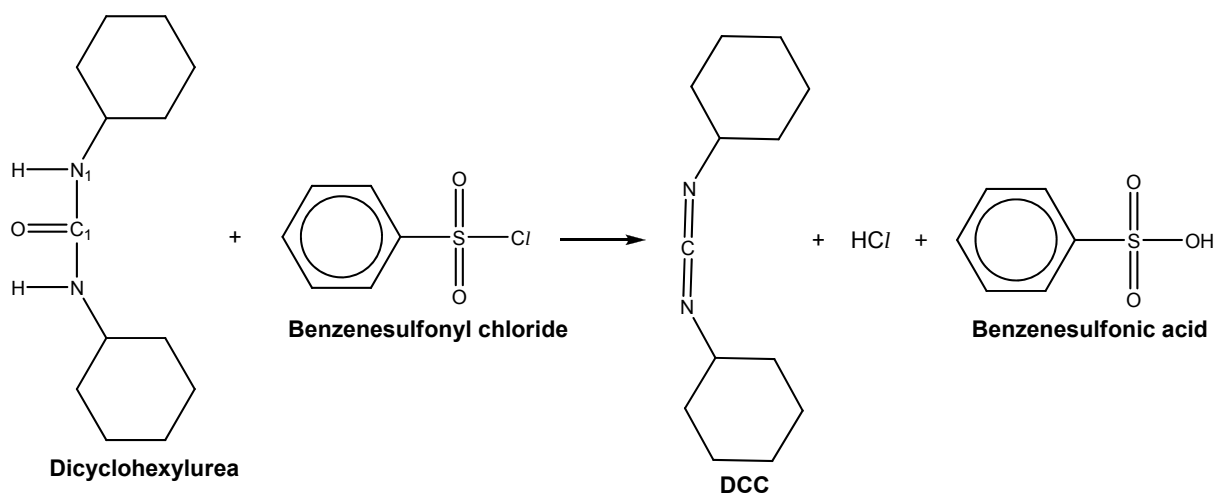
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.....[3]

[Turn Over

- 2 (c) DCC can also be synthesised from dicyclohexylurea and benzenesulfonyl chloride as shown in the following equation.



- (i) State the VSEPR (valence shell electron pair repulsion) theory to predict the shape about N₁ and C₁ in dicyclohexylurea.

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.....

[3]

- (ii) Draw the structural formulae of the products formed when dicyclohexylurea reacts with hot aqueous sodium hydroxide.

[2]

[Turn Over]

- 2 (c) (iii) Benzenesulfonic acid has a benzene ring. Describe the bonding in benzene in terms of orbital overlap. Draw relevant diagrams to illustrate your answer.

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[4]

- (iv) Weak acids typically have a K_a value that is less than 1. The pK_a value of benzenesulfonic acid is -2.8. Deduce if benzenesulfonic acid is a weak acid.

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.....

[1]

[Turn Over

- 2 (c) (v) Define the term conjugate base. Hence, draw the structural formula for the conjugate base of benzenesulfonic acid.

.....
.....

[2]

- (vi) Benzenesulfonic acid can be converted into benzenesulfonyl chloride by using a period 3 chloride such as phosphorus pentachloride. Two other chlorides of period 3 elements include silicon chloride and magnesium chloride.

Using the chlorides of magnesium, silicon and phosphorus as examples, describe the reactions of the chlorides of the third period of the Periodic table with water, writing equations where appropriate.

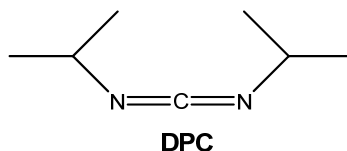
Hence, suggest what influence the type of bonding present in these chlorides has on their reaction with water.

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[5]

[Turn Over

- 2 (d) N,N-diisopropylcarbodiimide, DPC, was developed as an alternative to DCC. DPC is identical to DCC in nearly every way except that DPC is less likely to cause an allergic reaction. Furthermore, as a liquid, DPC is easier to handle than DCC which is a waxy solid. Explain the difference in physical states between DCC and DPC.



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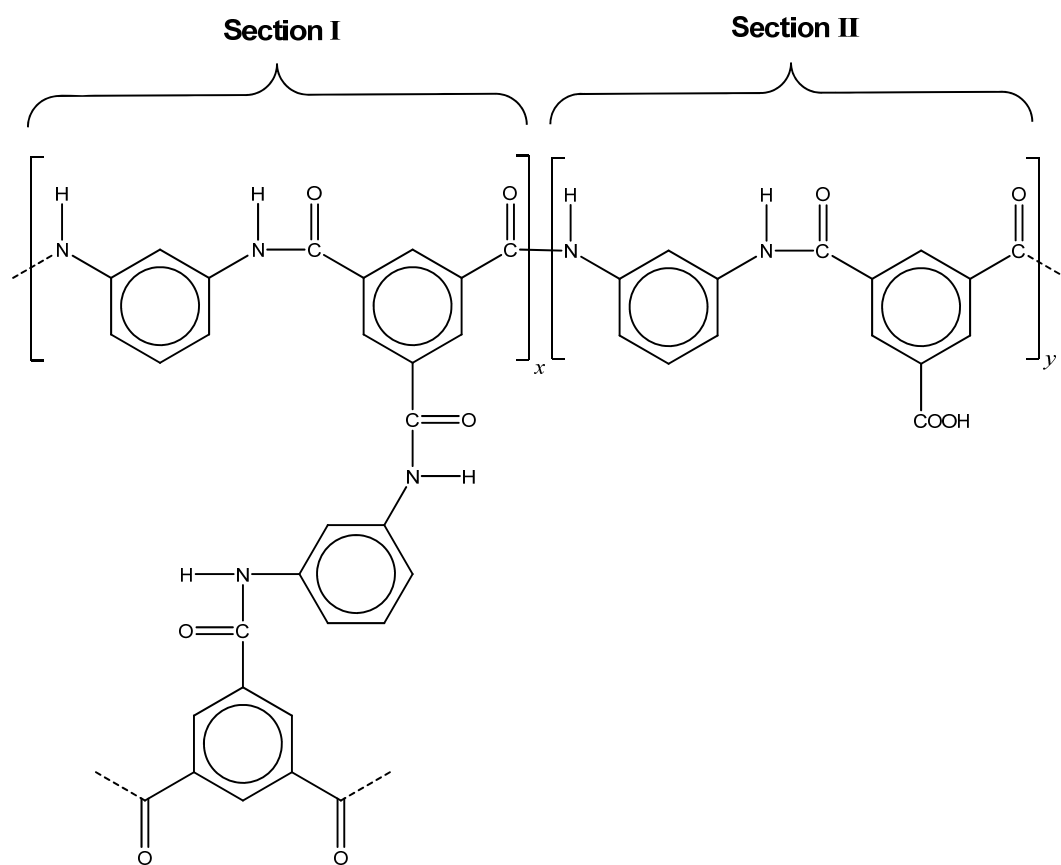
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[2]

[Total: 29]

[Turn Over

- 3 Nomex is a class of aramid fibres used as a heat-resistant and strong synthetic fibres in aerospace and military applications. Two sections of the polymer, Nomex, is as shown below.



- (a) Define polymer.

.....

.....

[1]

[Turn Over

- 3 (b) (i) With reference to Section II, draw the structural formula of the monomers in Nomex. Hence, state the type of polymerisation involved.

Monomers:

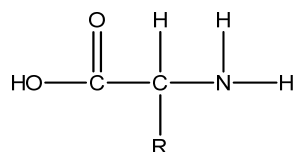
Type of polymerisation: [3]

- (ii) Classify the polymer represented by Section I. Hence, use its structure to explain a property of this polymer.

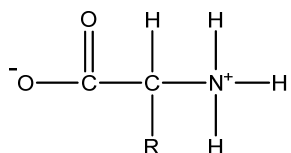
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[3]

- (c) Wool is a texture fibre obtained from sheep and other animals. Wool consists of a polymer chain constructed from amino acids. The general structure of an amino acid is as shown below.

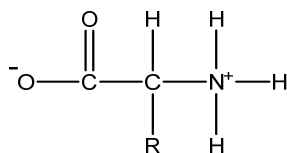


In water, amino acids forms zwitterions as shown below.



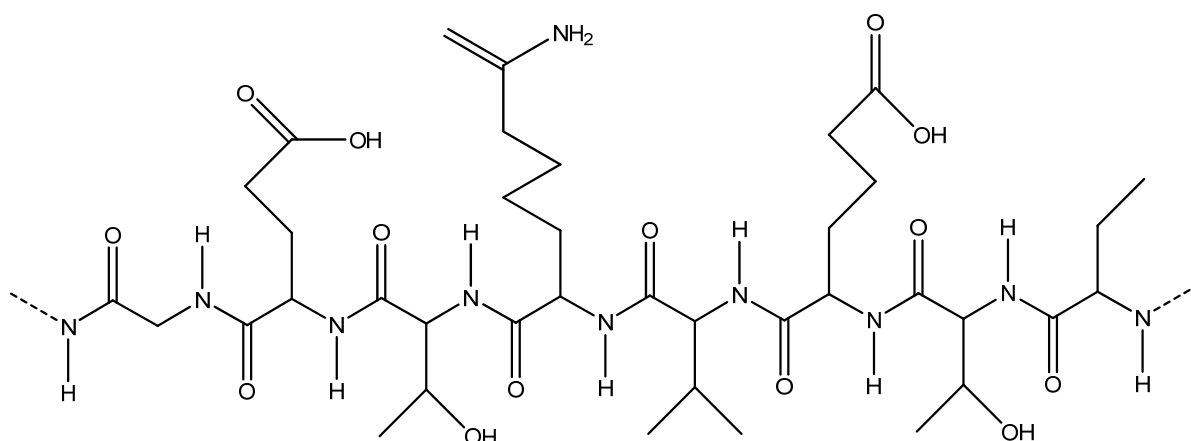
[Turn Over

- 3 (c) (i) Hygroscopy is the phenomenon of attracting moisture from the surrounding environment. Using the structure of zwitterion in the diagram below, show with the aid of a labelled diagram the hygroscopic nature of zwitterions.



[2]

The structure of a section of wool is shown below. This section consists of a number of amino acids.



- (ii) On the structure above, **circle** and **name** 4 different functional groups present in wool.

Functional groups:

.....

[2]

- (iii) How many amino acids are present in this portion of wool?

.....[1]

[Turn Over

- 3 (c) (iv) One of these amino acids are present more than once. Draw the displayed formula for this amino acid.

[1]

- (d) Silver nanoparticles are used to colour wool fibres as well as to impart antimicrobial properties. Define the term nanoparticles and explain with reference to its structure, the advantages of using silver nanoparticles.

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.....

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[2]

[Total: 15]

[Turn Over

Section B

Answer **one** question from this section, in the spaces provided.

- 1 (a) (i) State the type of bonding in magnesium and chlorine. Use a labelled diagram to illustrate each of your structure.

.....

.....

[4]

- (ii) Describe the structure of graphene.

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.....

[2]

- (iii) State the similarities in the structure of magnesium and graphene that allow both of them to conduct electricity. Hence, state another physical property that both magnesium and graphene will have in common.

.....

.....

[2]

[Turn Over

1 (b) Magnesium is an element in Period 3.

(i) Define the term nucleon number.

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.....

[1]

In a particular experiment set-up, protons are deflected through an angle of $+15^\circ$.

(ii) State one way in which the behaviour of electrons in an electric field differs from that of protons.

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.....

[1]

(iii) Calculate the angle of deflection of a Mg^+ particle.

[1]

(iv) Explain how the first ionisation energy of Mg would vary with the element before and after it.

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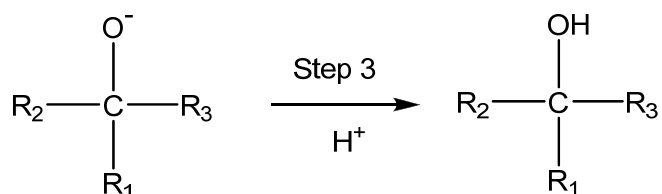
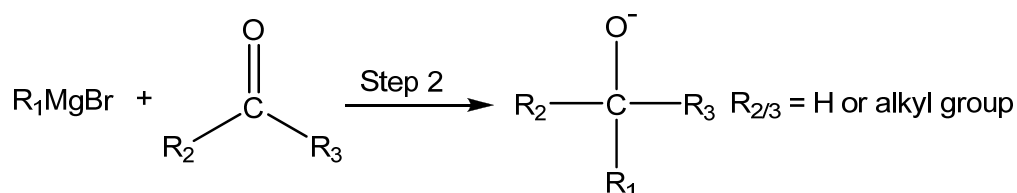
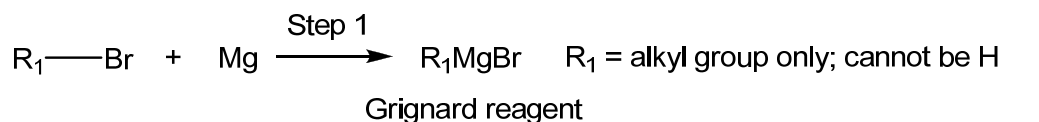
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[3]

[Turn Over]

- 1 (c) Grignard reagents are formed by the reaction of magnesium metal with an organic compound. They can then be used to react with a carbonyl compound to obtain an alcohol as the final product. This is known as the Grignard reaction and can be represented by the following reaction scheme.



- (i) Name the functional group of the organic compound that reacted with magnesium to form the Grignard reagent in Step 1.

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[1]

- (ii) Suggest the type of reaction that took place in Step 2.

.....

[1]

- 1 (c) (iii) Draw the structural formula and give the IUPAC name of the alcohol that is formed when bromoethane was used to form the Grignard reagent and subsequently reacted with butanal.

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[2]

- (iv) Butan-2-ol is an alcohol that can be obtained via the Grignard reaction. Draw the structure of a primary alcohol that is a constitutional isomer of butan-2-ol.

[1]

- (v) Suggest the functional group of the compound formed when carbon dioxide, CO_2 , is used in the Grignard reaction instead of the carbonyl compound.

.....
[1]

[Total: 20]

[Turn Over]

- 2 Aluminium and sulfur are elements found in Period 3. When burnt in excess oxygen, their oxides, Al_2O_3 and SO_3 , will be produced.

(a) Using relevant equations, justify the nature of both oxides.

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.....

.....

[3]

(b) (i) Draw the dot-and-cross diagram of SO_2 .

[1]

(ii) Hence or otherwise, deduce whether SO_2 or SO_3 has a smaller bond angle and explain your answer.

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.....

[1]

[Turn Over]

- 2 Sulfuric acid (H_2SO_4) is a commonly used chemical reagent in many industrial processes and is manufactured through the Contact process. This takes place in 3 stages.

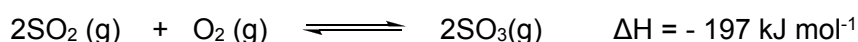
Stage 1:

In this stage, SO_2 is produced via burning pyrite, a compound containing only Fe and S, in excess oxygen. The equation at this stage is as follows:



Stage 2:

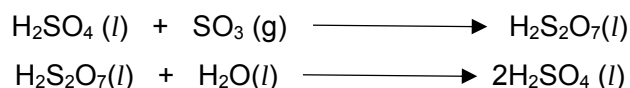
In this stage, the SO_2 produced is converted to SO_3 according to the following equation:



Stage 3:

In this stage, the SO_3 produced is converted to H_2SO_4 via 2 reactions. The first reaction involves dissolving the SO_3 produced in concentrated H_2SO_4 to obtain $\text{H}_2\text{S}_2\text{O}_7$.

The $\text{H}_2\text{S}_2\text{O}_7$ produced can then be safely added to water to produce H_2SO_4 in the second reaction. The equations at this stage are as follows:



To produce H_2SO_4 in a laboratory, a technician would first burn 10 g of pyrite in excess oxygen. The SO_2 produced would then be extracted and pumped into a 2 dm^3 vessel with more oxygen and allowed to react again at a moderately high temperature. When equilibrium has established, the SO_3 produced was quickly extracted and pumped into 50 cm^3 of concentrated H_2SO_4 before the resultant solution was added to 50 cm^3 of deionised water to produce H_2SO_4 .

The H_2SO_4 produced was tested for its concentration where 25.0 cm^3 of the resultant solution required 23.50 cm^3 of 2.00 mol dm^{-3} NaOH for complete titration.

- (c) (i) Given that pyrite contained 53.5% of sulfur by mass, show that the value of x in FeS_x is 2.

[1]

[Turn Over

2 (c) Using the information given above,

(ii) Calculate the number of moles of SO_2 produced from the combustion of pyrite.

[1]

(iii) Calculate the total number of moles of H_2SO_4 obtained by the technician and hence the number of moles of SO_3 produced.

[2]

(iv) Write an expression for the equilibrium constant, K_c , for the Contact process in **stage 2**.

[1]

[Turn Over]

- 2 (c) (v) Using your answers obtained in 2(c)(ii) – (iv), complete **Table 2.1**.

Table 2.1

	SO ₂	O ₂	SO ₃
initial amount/mol		1	
change/mol			
equilibrium amount/mol			

Hence or otherwise, calculate the value of K_c .

[2]

- (vi) To ensure a reasonable yield of SO₃, the technician conducted the reaction between SO₂ and O₂ at moderately high temperatures. Explain why this is so.

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[2]

[Turn Over

- 2 (d)** **X** has a molecular formula of $C_4H_{10}O_2$. 1 mole of **X** reacts with 1 mole of sodium to give 1 mole of hydrogen gas.

When acidified potassium manganate (VII) is heated with **X**, the solution turns colourless to give only **Y**.

When 1 mole **Y** is added to methanol in the presence of concentrated sulfuric acid, **Z** is obtained together with 2 moles of water.

Deduce the structures of **X**, **Y** and **Z**, explaining the chemistry of the reactions described.

[6]

[Total: 20]

END OF PAPER